ME 37500: Systems, Measurements, and Control II

Course Information

- **Meeting day(s) and time(s)**. Tuesday, Thursday
- **Instructional Modality.** Face-to-Face.
- **Course credit hours.** 3
- **Prerequisites.** ME 36500: Systems, Measurements, and Control I, MA 26500: Linear Algebra

Instructor(s) Contact Information

Instructor	E-mail address	Lecture day / time / room
Neera Jain (course supervisor)	neerajain@purdue.edu	T/R | 9:30am-10:20am | ARMS B061
Todd Lillian	tlillian@purdue.edu	T/R | 10:30am-11:20am | WALC 2087
		T/R | 12:30pm-1:20pm | ARMS 1010
Song Zhang	zhan2053@purdue.edu	T/R | 1:30pm-2:20pm | ME 1130
Raghav Kakani	kakani@purdue.edu	T/R | 4:30pm-5:20pm | ME 1130

We have two head TAs for this course:

- Lead Lecture TA: Falak Mandali (fmandali@purdue.edu)
- Lead Laboratory TA: José Ernesto Sandoval Hernandez (sandov18@purdue.edu)

A complete list of the graduate and undergraduate teaching assistants (TAs), along with their contact information, is available on Brightspace.

For questions or concerns specific to you, please use e-mail to contact your instructor.

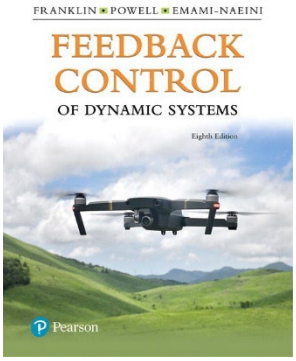
For general administrative, homework, laboratory, or exam questions, please use Brightspace discussion boards or attend office hours. If you e-mail an instructor or TA with a question whose answer is relevant to the entire class, you will be asked to post the question on the discussion board instead.

Course Description

This course builds on the measurement, system modeling, and system/analysis fundamentals introduced in ME 36500 (Systems, Measurements and Control I), with an emphasis on closed-loop feedback control theory and design. The mathematical tools necessary for analyzing closed-loop stability and performance (e.g. steady-state error, transient performance characteristics) are presented. Pole placement, root locus, and frequency domain methods are used to design feedback control algorithms. Hands-on assignments are utilized to reinforce fundamental systems, measurement, and control concepts, including the implementation of control algorithms in simulation and on physical hardware.

Learning Resources, Technology & Texts

In ME 375, we will be using the following required text:



Franklin, Gene F., J. David Powell, and Abbas Emami-Naeini. *Feedback control of dynamic systems*. Upper Saddle River, NJ: Pearson, 2019.

This citation is specifically the US 8th edition of the textbook—NOT the global edition. You are strongly encouraged to rent or buy this version as homework problems will be assigned from this edition. A picture of the cover is shown to the right for your reference.

A hard copy of the textbook is available for purchase at the **University Bookstore.** The **e-book can be rented or purchased online here**: https://www.pearson.com/en-us/pearsonplus/p/9780137516834

A copy has also been placed **on reserve in the Engineering Library in WALC** if you are unable to rent or purchase the text.

Technology

- ***Brightspace*** learning management system (LMS) will be used for dissemination of lecture material, assignments, and announcements, as well as to field administrative and homework questions via discussion boards.
- ***Gradescope*** will be used for submission and grading of homework and laboratory assignments.
- ***iClicker*** will be used in lecture for checking understanding but will not count towards your final grade.
- ***MATLAB® and Simulink®*** will be used on homework assignments and in the laboratory.
- ***Texas Instruments or Casio calculator*** satisfying the ME Exam Calculator Policy.

Policy on use of Artificial Intelligence

In general, Artificial intelligence (AI) or Large Language Models (LLM) **should not be used to complete any homework or laboratory assignments**. However, you may use a LLM to clarify your understanding of a particular concept. **You are *strongly encouraged* to prioritize using the textbook, lecture notes, and your instructor or TA to learn the concepts presented in this course.**

Tutoring Support

Instructors and TAs will schedule weekly office hours to be held throughout the semester. See Brightspace for the most up-to-date schedule. You are also strongly encouraged to use Brightspace discussion boards to seek help; the discussion boards will be regularly monitored by the instructional team.

Learning Outcomes

1. Students will understand the mathematical theory governing closed-loop system stability and performance.
2. Students will learn how to design single-input-single-output linear feedback control algorithms to achieve closed-loop stability and specified system performance objectives.
3. Students will learn how to implement control algorithms on physical hardware.
4. Students will improve their technical communication skills through laboratory assignments.

Assignments and Grading

Your achievement of course learning outcomes will be assessed through a combination of participation, homework assignments, laboratory assignments, midterm exams, and a final exam spread throughout the academic period. Details on these assignments and exams, including a schedule of due dates will be posted on the course website.

Table 1: Grading Overview

Assignments	Due	Percentage of Total Grade
Homework	Weekly: Thursday at 9pm ET	5%
Laboratory	Weekly: 9pm ET on day of your laboratory session	30%
Midterm Exam 1	February 24, 2026 (6:30-7:30pm ET)	20%
Midterm Exam 2	April 14, 2026 (6:30-7:30pm ET)	20%
Final Exam	TBD	25%
		Total: 100%

- **HOMEWORK (5%; weekly).** Homework will be posted to Brightspace and Gradescope on a weekly basis. Once you have completed each homework assignment, scan and upload your completed assignment to Gradescope (gradescope.com) for grading. You will receive notification of a Gradescope account (using your Purdue e-mail address) at the beginning of the semester. A guide for scanning and submitting homework is available on the Gradescope web site.
  - **No late homework will be accepted, and no exceptions will be made to this policy. Instead, your *lowest two homework assignment grades will be dropped at the end of the semester.***
  - Retain all records of your Gradescope submissions (i.e., keep e-mails verifying successful submissions to Gradescope).
  - Solutions to the homework will be available on Brightspace after the assignment is due.
  - Homework will be graded for completeness and general approach to the assigned problems.
  - Homework **must be neatly hand-written and scanned clearly (in Gradescope)** in addition to being correct to obtain full credit. If a grader cannot interpret your solution, you will receive no credit.
  - Start each problem on a new sheet of paper.

- **LABORATORY (30%; weekly).** In each of the laboratory sessions, you will complete a MATLAB® Live Script file to be submitted to your lab instructor no later than 9pm ET on the day of your lab section. There will be no pre- or post-laboratory work (besides submitting your assignment by 9pm ET that day).
  - **Attendance**
    - Lab begins on time; if you arrive more than 15 minutes late, you will receive a zero.
      - You are still encouraged to stay and complete the lab assignment since each lab builds on the previous one.
    - Only students with absences verified by the Office of the Dean of Students or excused by the instructional team will be allowed to make up a missed laboratory session. In this case, contact your lab instructor, the lead lab TA (sandov18@purdue.edu), and the lead course instructor (neerajain@purdue.edu) to reschedule your lab either by attending a different section that week or scheduling a time during a dedicated make-up week in the semester.
    - Any unexcused absence will result in a zero grade for that laboratory session.
  - **Compliance with laboratory guidelines**
    - You are responsible for following all specified rules and safety guidelines in ME 1036/1042. If after one warning, noncompliant or unsafe behavior persists, you will be asked to leave and will receive a zero grade for that laboratory session.

Regrade requests for homework and laboratory assignments will be accepted for one week after grades are published on Gradescope.

- **TWO MIDTERM EXAMS (40%; see dates in Table 1).** Your learning will be assessed using two multiple-choice midterm exams distributed across the semester. The dates are provided in Table 1.

- **FINAL EXAM (25%).** The final exam will consist of multiple-choice questions and be cumulative in its coverage of the course material.

Grading Scale

In this class, grades reflect the sum of your achievement of learning outcomes throughout the semester. At the end of the semester, final grades will be calculated by computing your percentage grade for each group of assignments and then weighing each assignment type according to Table 1. The weighted percentage will be translated into a letter grade according to the scheme presented in Table 2. Numerical grades will be rounded to the nearest whole number using a standard rounding scheme; for example, 93.5% would be rounded to 94%; 93.4% would be rounded down to 93%.

Table 2: Grading Scale

Percentage	Letter Grade
97-100%	A+
93-96%	A
90-92%	A-
87-89%	B+
83-86%	B
80-82	B-
77-79	C+
73-76	C
70-72	C-
67-69	D+
63-66	D
60-62	D-
< 59 %	F

How to Succeed in this Course

If you want to be a successful student in ME 375:

- Be self-motivated and self-disciplined.
- Be willing to speak up if problems arise.
- Access the resources that are available to you; reaching out for assistance is a critical life skill rather than an indication of weakness.
- Be willing and able to commit to approximately 5-10 hours per week of time outside the course on homework and study.
- Be able to communicate through writing.
- Be able to meet the minimum requirements for the course.
- Accept critical thinking as part of the learning process.
- Take advantage of resources such as those listed in our course Brightspace under Announcements regarding Technology, Academics, Campus Resources, etc.
- Review the University Policies content on our course Brightspace.

Attendance Policy

This course follows the Academic Regulations: Attendance and Office of the Dean of Students: Class Absences posted in Brightspace under "University Policies and Statements." The policies state that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance will be taken at the beginning of each class and lateness will be noted. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform me of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to is not possible, contact me as soon as possible by email or phone.

For cases that fall under the University excused absence regulations – Grief/Bereavement, Military Service, Jury Duty, Parenting Leave, or the Medically Excused Absence Policy for Students -- you or your representative go to the Office of the Dean of Students (ODOS) website to complete appropriate request forms. ODOS reviews these requests and, if granted, will notify all your instructors.

Course Schedule

An up-to-date course schedule is available on Brightspace.

Course Evaluation

Toward the end of this course, you will be provided with an opportunity to give feedback on the course and your instructor. Purdue uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site and will receive a prompt to complete the survey when you login to Brightspace. The subject line will be: *Please take 2-5 minutes to complete the survey*. Check your "Junk E-mail" folder occasionally to be sure the evaluation emails were not accidentally routed there. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

Purdue's student conduct regulations state the following regarding academic integrity:

"Students are expected to adhere to the guidelines provided by instructors for academic work so that no student gains an unfair advantage. Using or attempting to use unauthorized materials, information, study aids, notes, or any other device in any academic exercise will not be tolerated. Unauthorized materials may include anything which or anyone who gives a student assistance that has not been approved by the instructor in advance."

All assignments are to be completed independently, except for laboratory assignments, which will be completed in coordination with a partner. Plagiarizing all or part of any of these assignments will result in the student being reported to the Office of Student Rights and Responsibilities (OSRR) for review at the university level and a grade penalty will be assessed of up to and including a failing grade in the class.

Nondiscrimination Statement

Purdue University and I are committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages everyone to strive to reach their potential. For details, see Purdue's full Nondiscrimination Policy Statement.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. The Disability Resource Center (DRC) is a key partner in this work and is a resource for students and instructors. A link to the DRC website is located on the Brightspace landing page.

Mental Health/Wellness Statement

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. The Community, Assistance and Resources for Engineering Students (CARES) Hub is open to support the well-being of all engineering students. The CARES Hub, located in ARMS 1261, is a welcoming, inclusive space for students to study, connect, grab a snack, and relax 7 days per week. There will also be wellness activities, educational workshops, peer mentoring, social events, and on-site counseling. If you need help handling stress or working through a problem, schedule time with our CARES Hub therapist, Jennie Beutler, here or drop by ARMS 1251 to speak to Jennie without an appointment during her **Let's Talk** walk-in hours from 12:30-1:30pm Monday-Friday. Please visit the CARES Hub webpage to learn more. The CARES Hub staff look forward to connecting with you!

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of CAPS**.** TAO is available to you at any

time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression, and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

Student- and Safety-Related Policies

Purdue University provides numerous policies related to students. A hyperlink to this list of Student-Related Policies is on the Brightspace landing page. I suggest you review these policies, particularly those under Ethics and Facilities and Safety. For example, see the Violent Behavior Policy that focuses on our efforts to provide safe and secure campus environments for members of the university community. Therefore, violent behavior is prohibited in or on any University facility or while participating in any university activity. Another example is the policy prohibiting Amorous Relationships between a student and any University employee who has educational responsibility over the student. These and similar policies are in place as part of Purdue's commitment to maintaining an environment in which learning, discovery, and engagement take place in safe and professional environments.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Disclaimer

This syllabus is subject to change. You will be notified of any changes as far in advance as possible via an announcement on Brightspace. Monitor your Purdue email daily for updates.